

Bilateral Drive Gear—A Highly Backdrivable Reduction Gearbox for Robotic Actuators

Hiroshi Matsuki , Kenta Nagano , *Member, IEEE*, and Yasutaka Fujimoto , *Senior Member, IEEE*

Abstract—For safety purposes, cooperative robots are installed with an actuator composed of a low-power servo motor, a reduction gearbox, and a torque sensor. When cooperative robots make contact with humans or the environment, they must detect the contact force with a force sensor, a contact sensor, or a joint torque sensor. Equipping these sensors increases the cost and size of the application, but can be avoided under sufficient backdrivability of the actuator. To this end, we propose a method that maximizes the power transmission efficiency of the 3K planetary reduction gearbox and develop a prototype of the backdrivable reduction gearbox called the bilateral drive gear. For this maximization, the profile shift coefficients and the number of teeth are decided under some conditions. The forward- and backward-driving efficiencies of the prototype gearbox were 89.0% and 85.3%, respectively, and the reverse-drive starting torque was 0.020 N·m. The drive efficiency of the same gearbox with uncorrected teeth is 68.5%. The forward-driving efficiency was 20.5% higher than the nonoptimized one. We confirmed that prototype gearboxes with different gear ratios are easily backdrivable by hand.

Index Terms—Compound planetary reduction gearbox, cooperative robot, robotic actuator.

I. INTRODUCTION

UNDER appropriate risk assessment, cooperative robots that can operate immediately next to workspaces without a safety fence are being actively developed in many fields [1]. For instance, mechatronic systems have been developed for agricultural [2], [3] and industrial [4], [5] tasks. For safety purposes, cooperative robots have an actuator composed of a low-power servo motor, a reduction gearbox, and a torque sensor. Exoskeleton assist applications have also been developed in many institutions. Examples are lower-limb wearable robots [6],

[7], wearable robots for carrying heavy loads [8], upper extremity exoskeleton robots [9], flexible wearable robots that operate by jamming of granular media [10], and exoskeletons for neurorehabilitation [11]. Assist robots employ geared motors, hydraulic actuators, or pneumatic actuators [12].

A single motor in these robots cannot generate sufficient torque given the weight of the apparatus. Therefore, the motor is combined with a reduction gearbox with a gear ratio between 1/10 and 1/300 [13], [14]. As the reduction ratio increases, the reverse drivability (backdrivability) of the actuator decreases because these actuation systems have more high-impedance transmissions. Therefore, when deployed in contact situations with humans or the environment, the robot must detect the contact using a force sensor, a contact sensor, or a joint torque sensor.

To improve the reverse driving performance of an actuator with speed reduction, researchers have developed a series elastic actuator (SEA) [15]–[18] and actuators with a built-in torque sensor [19]–[21]. The SEA includes a low-rigidity elastic element between the output of the speed reducer and the load, which measures the displacement under the load. While this configuration realizes accurate torque control (i.e., good force control), it provides poor positioning accuracy. In addition, the SEA cannot be passively backdriven when unpowered. In an actuator with a built-in torque sensor, the performance of the torque control depends on the sensor band, and the control band is limited; moreover, the safety at the time of sensor failure is not guaranteed. Both the SEA and built-in torque-sensing methods require additional elastic elements and sensors, which increase their cost. Another type of actuator employing a low-ratio gearbox has been developed to reduce the rotor apparent inertia and to improve the actuator bandwidth by increasing the torque density of the actuator and decreasing the transmission ratio [22], [23]. Reduction of the rotor apparent inertia is important for human-safe robots in order to minimize the impact force.

We are developing an actuator that detects forces and contacts by increasing the reverse drivability of the reducer itself, negating the need for additional elastic elements and sensors [24], [25]. Table I summarizes the characteristics of various speed reducers installed in robot joints. As a simple planetary gear reducer cannot have a large reduction ratio, it is often connected through a plurality of stages. In this article, we investigate the 3K planetary gearbox, in other words, Wolfrom gearbox, which is one of the compound planetary gearboxes. In general, this gearbox has advantages of the compact structure and a large reduction ratio [26], [27]. However, this efficiency decreases

Manuscript received October 15, 2018; revised April 30, 2019 and August 23, 2019; accepted September 30, 2019. Date of publication October 14, 2019; date of current version December 31, 2019. Recommended by Technical Editor J. Yoon. This work was supported by the New Energy and Industrial Technology Development Organization of Japan. (Corresponding author: Yasutaka Fujimoto.)

H. Matsuki was with the Department of Electrical and Computer Engineering, Yokohama National University, Yokohama 240-8501, Japan. He is now with Ashikaga University, Ashikaga 326-8558, Japan (e-mail: matsuki.hiroshi.rs@gmail.com).

K. Nagano and Y. Fujimoto are with the Department of Electrical and Computer Engineering, Yokohama National University, Yokohama 240-8501, Japan (e-mail: nagano-kenta-cw@ynu.ac.jp; fujimoto@ynu.ac.jp).

This article has supplementary downloadable material available at <http://ieeexplore.ieee.org>, provided by the authors.

Color versions of one or more of the figures in this article are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TMECH.2019.2946403

TABLE I
CHARACTERISTICS OF GEARBOXES USED IN ROBOTS

Type of gearbox	Reduction ratio	Torque capacity	Size	Efficiency (%)	Precision (arcmin)
Simple planetary[34]	1/4 – 1/10	low	small	>95	>3
Two-stage planetary[34]	1/10 – 1/50	low	small	<90	>3
Harmonic[34]	1/30 – 1/160	mid	mid	60 – 80	1 – 3
Simple cycloid[35]	1/5 – 1/20	high	big	>95	<1
Two-stage cycloid[35]	1/30 – 1/300	high	big	<90	<1
Planetary+cycloid[36]	1/30 – 1/200	high	big	<90	<1

according to the increase in the reduction ratio [28], [29]. On the other hand, harmonic gear reducers are lightweight, compact, and have a large reduction ratio with high precision but low efficiency. Cycloid gear reducers deliver high torque capacity with both high accuracy and high efficiency. Mechanical paradox gearboxes usually have low-efficiency characteristics between 70% and 75%. Hori proposed efficiency improvement of the mechanical paradox gearbox in [13]. He achieved up to 80–85% efficiency. In general, there are several tradeoffs among the power transmission efficiency, reduction ratio, allowable torque, size, and backlash of the speed reducer [30], [31]. More specifically, reducing the reduction ratio increases the permissible torque, whereas reducing the size or backlash of the reducer decreases the power transmission efficiency. In fact, reverse driving by a small speed reducer with a reduction ratio of $\sim 1/100$ is difficult to achieve in conventional robot applications.

There are some papers that refer to the gearbox. Wang and Kim explain the backdrivability of an actuator with a gearbox by investigating the directionality of the efficiency of the spur gears [32]. This article shows that the driving efficiency in 6:1 gearbox is 2% higher than the backdriving efficiency. Deb and Jain optimized the design method of the multispeed gearbox using a multiobjective evolutionary algorithm [33]. By comparison with these papers, we achieved the high backdrivability and the high driving efficiency in the 3K planetary gearbox with a high gear ratio by optimizing the total efficiency using the profile shift and the number of teeth.

Herein, we propose a method that maximizes the power transmission efficiency of the 3K planetary gearbox. Prototype speed reducers with reduction ratios of 1/50–1/380 are easily operated in reverse drive, and the starting torque at the time of reverse driving is also extremely small. The power transmission efficiency of the 1/100 prototype speed reducer is approximately 89.0%.

II. GENERAL MODEL OF REDUCTION GEARBOXES

This section derives the general relationship between the reduction ratio and the driving efficiency in single-stage high-reduction gearboxes. The model approximates general gearboxes made of a combination of a pair of gears. The general model that considers the friction at the sliding surface is schematized in Fig. 1. This figure illustrates the general but minimal model of force transmission and velocity transformation. We introduce the backdrivability, the driving efficiency, and the backdriving efficiency using this model.

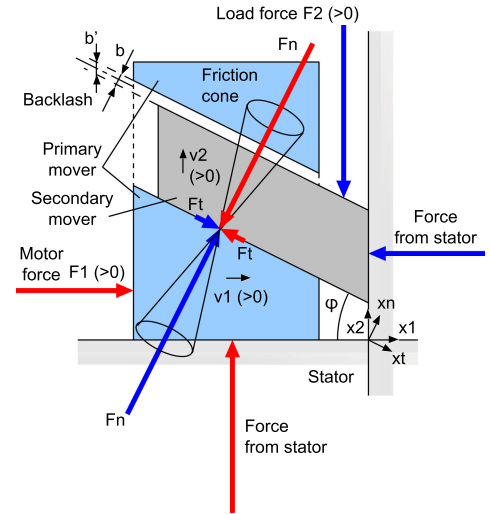


Fig. 1. General model of reduction gearboxes; red and blue arrows show the forces acting on the primary and secondary movers, respectively. The variables are defined in the text.

The model consists of three components: a primary mover, a secondary mover, and a stator. The horizontally moving primary mover may contact the vertically moving secondary mover on the smooth sliding surface. The positions of the primary and secondary movers are represented by x_1 and x_2 , respectively. The tangential and normal components of the displacement of the secondary mover relative to the primary mover are, respectively, given by

$$x_t = -x_1 \cos \phi - x_2 \sin \phi \quad (1)$$

$$x_n = -x_1 \sin \phi + x_2 \cos \phi \quad (2)$$

and their derivatives are

$$v_t = \dot{x}_t = -v_1 \cos \phi - v_2 \sin \phi \quad (3)$$

$$v_n = \dot{x}_n = -v_1 \sin \phi + v_2 \cos \phi \quad (4)$$

where $v_1 = \dot{x}_1$ and $v_2 = \dot{x}_2$. Assume a small friction on the sliding surface and a small backlash between the primary and secondary movers. When the gearbox transmits a force, the primary mover contacts the secondary mover and $v_n = 0$. In this case, $v_2 = v_1 \tan \phi$, and the inverse of the reduction ratio of the gearbox is given by

$$G = \tan \phi \quad (5)$$

noting that $G = \tan \phi \ll 1$, which indicates a large transmission reduction. Fig. 1 illustrates the basic principle of the kinetic force transmission from the motor-side primary mover to the load-side secondary mover. The motor force F_1 that horizontally pushes the primary mover is balanced by the sum of the normal force F_n , the friction force F_t received from the sliding surface, and the force received from the stator. On the sliding surface, the normal and friction forces on the secondary mover are of equal magnitude but opposite directions. They are balanced by the load force F_2 and the force from the stator. The mover dynamics are formulated in detail in the following.

Referring to Fig. 1, the motion equations of the primary mover and secondary movers are, respectively, given by

$$M_1 \dot{v}_1 = F_1 - F_n \sin \phi - F_t \cos \phi \quad (6)$$

$$M_2 \dot{v}_2 = -F_2 + F_n \cos \phi - F_t \sin \phi \quad (7)$$

where M_1 and M_2 are the masses of the primary and secondary movers, respectively, F_1 is the motor force acting on the primary mover, and F_2 is the load force imposed on the secondary mover in the negative direction. F_n and F_t are the normal and tangential forces acting on the sliding surface, respectively. The tangential force F_t is given by

$$F_t = -\mu' |F_n| \text{sgn}(v_t) \quad (8)$$

where μ' is the friction coefficient of the sliding surface. We assume that μ' does not exceed the inverse of the reduction ratio G , i.e.,

$$\mu' \leq G. \quad (9)$$

Owing to backlash between the primary and secondary movers, the bottom and top surfaces of the secondary mover contact the primary mover when $x_n = -b/2$ and $x_n = b/2$, respectively, where b is the maximum gap between the primary and secondary movers.

A. Contact Mode at the Bottom Surface ($x_n = -b/2$)

When the primary mover contacts the bottom surface of the secondary mover, we have $v_n = 0$, and (6) and (7) can be rewritten as

$$M_1 \dot{v}_1 = F_1 - F_n (\sin \phi - \text{sgn}(v_t) \mu' \cos \phi) \quad (10)$$

$$M_2 \dot{v}_2 = -F_2 + F_n (\cos \phi + \text{sgn}(v_t) \mu' \sin \phi) \quad (11)$$

$$F_n \geq 0 \quad (12)$$

$$v_2 = G v_1 \quad (13)$$

$$v_t = -\frac{v_1}{\cos \phi} = -\frac{v_2}{\sin \phi}. \quad (14)$$

Eliminating F_n from (10) and (11), we obtain

$$F_1 - M_1 \dot{v}_1 = \frac{G - \text{sgn}(v_t) \mu'}{1 + \text{sgn}(v_t) \mu' G} (F_2 + M_2 \dot{v}_2). \quad (15)$$

Note that $F_1 - M_1 \dot{v}_1 \geq 0$ and (10)–(12) give $F_2 + M_2 \dot{v}_2 \geq 0$. Multiplication by v_1 converts the motion equation (15) into the following:

$$\begin{aligned} F_1 v_1 - \frac{d}{dt} \left(\frac{1}{2} M_1 v_1^2 \right) \\ = \frac{G - \text{sgn}(v_t) \mu'}{(1 + \text{sgn}(v_t) \mu' G) G} \left(F_2 v_2 + \frac{d}{dt} \left(\frac{1}{2} M_2 v_2^2 \right) \right). \end{aligned} \quad (16)$$

This equation describes the power flow of the reduction gear, which depends on the sign of v_t . The forward- and backward-driving efficiencies and the motion equation are described as follows.

1) Forward Driving: During forward driving, $v_t \leq 0$, i.e., $v_1 \geq 0$ and $v_2 \geq 0$. Under these conditions, (16) becomes

$$F_2 v_2 + \frac{d}{dt} \left(\frac{1}{2} M_2 v_2^2 \right) = \eta \left(F_1 v_1 - \frac{d}{dt} \left(\frac{1}{2} M_1 v_1^2 \right) \right) \quad (17)$$

where the forward-driving efficiency η is given by

$$\eta = \frac{(1 - \mu' G) G}{G + \mu'} \simeq \frac{1}{1 + \mu' / G}. \quad (18)$$

In this case, part of the input power $F_1 v_1$ is converted to kinetic energy, and the rest is transferred to the secondary mover. At high-reduction ratios, the efficiency of the power transmission is a hyperbolically decreasing function of the reduction ratio.

The power balance equation (17) can be rewritten in terms of the velocity relationship (13) as

$$F_2 v_2 + \frac{d}{dt} \left(\frac{1}{2} \tilde{M}_2^+ v_2^2 \right) = \eta F_1 v_1 \quad (19)$$

and

$$F_2 v_2 = \eta \left(F_1 v_1 - \frac{d}{dt} \left(\frac{1}{2} \tilde{M}_1^+ v_1^2 \right) \right) \quad (20)$$

where $\tilde{M}_1^+$ and $\tilde{M}_2^+$ are the equivalent masses in the primary and secondary coordinates, respectively, during forward driving:

$$\tilde{M}_1^+ = M_1 + \frac{G^2}{\eta} M_2 \quad (21)$$

$$\tilde{M}_2^+ = \frac{\eta}{G^2} M_1 + M_2 = \frac{\eta}{G^2} \tilde{M}_1^+. \quad (22)$$

Equation (22) shows that the equivalent mass in the secondary coordinates is inversely proportional to the square of the reduction ratio compared to the equivalent mass in the primary coordinates. Both equivalent masses are also affected by the forward-driving efficiency η .

In terms of the above equivalent masses, the motion equation can be derived as follows. Using (13), the original (15) can be rewritten in the forward-driving mode ($v_1, v_2 \geq 0$) as

$$\tilde{M}_1^+ \dot{v}_1 = F_1 - \frac{G}{\eta} F_2. \quad (23)$$

In secondary coordinates, the original (15) during forward driving becomes

$$\tilde{M}_2^+ \dot{v}_2 = \frac{\eta}{G} F_1 - F_2. \quad (24)$$

2) Backward Driving: During backward driving, we have $v_t \geq 0$, i.e., $v_1 \leq 0$ and $v_2 \leq 0$. Therefore, the power balance equation (16) becomes

$$F_1 v_1 - \frac{d}{dt} \left(\frac{1}{2} M_1 v_1^2 \right) = \eta' \left(F_2 v_2 + \frac{d}{dt} \left(\frac{1}{2} M_2 v_2^2 \right) \right) \quad (25)$$

where the backward-driving efficiency η' is defined as

$$\eta' = \frac{G - \mu'}{(1 + \mu' G) G} \simeq 1 - \mu' / G. \quad (26)$$

The power from the secondary mover is transferred to the primary mover. At high-reduction ratios, the efficiency of the backdriving power transmission becomes a linearly decreasing

function of the reduction ratio. When the friction coefficient μ' equals the reduction ratio $\tan \phi$, the backward-driving efficiency becomes zero, and the gearbox is no longer backdrivable.

This power balance equation can be rewritten in terms of the equivalent masses, as described in the forward-driving case

$$F_1 v_1 - \frac{d}{dt} \left(\frac{1}{2} \tilde{M}_1^- v_1^2 \right) = \eta' F_2 v_2 \quad (27)$$

$$F_1 v_1 = \eta' \left(F_2 v_2 + \frac{d}{dt} \left(\frac{1}{2} \tilde{M}_2^- v_2^2 \right) \right) \quad (28)$$

where

$$\tilde{M}_1^- = M_1 + \eta' G^2 M_2 \quad (29)$$

$$\tilde{M}_2^- = \frac{1}{\eta' G^2} M_1 + M_2 = \frac{1}{\eta' G^2} \tilde{M}_1^-. \quad (30)$$

The equivalent masses are also affected by the backward-driving efficiency η' .

In the backward-driving mode ($v_1, v_2 \leq 0$), the motion equations in the primary and secondary coordinates are, respectively, given by

$$\tilde{M}_1^- \dot{v}_1 = F_1 - \eta' G F_2 \quad (31)$$

and

$$\tilde{M}_2^- \dot{v}_2 = \frac{1}{\eta' G} F_1 - F_2. \quad (32)$$

B. Contact Mode at the Top Surface ($x_n = b/2$)

When the primary mover contacts the top surface of the secondary mover, (6) and (7) can be rewritten as

$$M_1 \dot{v}_1 = F_1 - F_n (\sin \phi + \text{sgn}(v_t) \mu' \cos \phi) \quad (33)$$

$$M_2 \dot{v}_2 = -F_2 + F_n (\cos \phi - \text{sgn}(v_t) \mu' \sin \phi) \quad (34)$$

$$F_n \leq 0. \quad (35)$$

Similarly to Section II-A, (13) and (14) are also satisfied. From (33)–(35), we obtain $F_1 - M_1 \dot{v}_1 \leq 0$ and $F_2 + M_2 \dot{v}_2 \leq 0$.

1) **Forward Driving:** The forward-driving condition is $v_t \geq 0$. Although this condition differs from the case in Section II-A, it leads to the same equations (17)–(24).

2) **Backward Driving:** The backdriving condition is $v_t \leq 0$. Under this condition, we can rederive (25)–(32).

C. Noncontact Mode ($-b/2 < x_n < b/2$)

The motion equations in the noncontact mode are simply described by

$$M_1 \dot{v}_1 = F_1 \quad (36)$$

$$M_2 \dot{v}_2 = -F_2. \quad (37)$$

When fabricating actual reduction gears, the backlash b is adjusted to be as small as possible. This mode exerts almost negligible effect in the efficiency analysis.

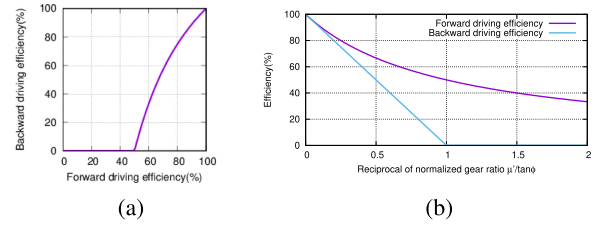


Fig. 2. Efficiencies of general reduction gears. (a) Backward- versus forward-driving efficiency derived from (39). (b) Forward- and backward-driving efficiencies versus reciprocal of the normalized gear ratio, derived from (18) and (26), when μ' is 0.01.

D. Backward- and Forward-Driving Efficiencies

Eliminating μ' from (18) and (26), we have

$$\eta' = \begin{cases} \frac{2\eta - 1 + G^2}{(1 - G^2)\eta + 2G^2}, & \left(\text{if } \frac{1 - G^2}{2} < \eta \leq 1 \right) \\ 0, & \left(\text{if } 0 \leq \eta \leq \frac{1 - G^2}{2} \right) \end{cases} \quad (38)$$

Under condition $G \ll 1$, (38) can be approximated as follows:

$$\eta' \approx \begin{cases} 2 - \frac{1}{\eta}, & \left(\text{if } \frac{1}{2} < \eta \leq 1 \right) \\ 0, & \left(\text{if } 0 \leq \eta \leq \frac{1}{2} \right) \end{cases} \quad (39)$$

These results reveal the following.

- 1) Improving the forward-driving efficiency also improves the backward-driving efficiency [see Fig. 2(a)].
- 2) The forward- and backward-driving efficiencies decrease hyperbolically and linearly, respectively, with increasing μ'/G (the quotient of the equivalent friction coefficient and the reduction ratio) [see Fig. 2(b)]. High-reduction-ratio gearboxes tend to be nonbackdrivable; the backdriving efficiency is zero when $G < \mu'$.

The equivalent friction coefficient μ' depends on the structure of the reduction gearbox. This friction corresponds to the friction occurring on the surface of tooth when a gear is sliding. A large reduction ratio with a high driving efficiency requires a structure with a small equivalent friction coefficient.

III. TYPE-3K COMPOUND PLANETARY GEARBOXES

This section describes the 3K compound planetary gearbox shown in Fig. 3. In our design method, we formulated the forward-driving efficiency of this reduction gearbox. The combination of teeth numbers and the profile shift coefficients of the teeth that maximize the forward-driving efficiency were determined by an optimization algorithm.

In this article, the gearbox shown in Fig. 4 is defined as 3K. The 3K planetary gear is one of the planetary gearboxes. This gearbox is called Wolfrom gearbox too. It has a drive body, a driven body, two axial adjacent ring gears, and the planetary rods. The main rods are composed of three rods of the spur gear (K means this rod). Fig. 4 shows a stick diagram of the 3K compound planetary gearbox shown in Fig. 3. The first planet gear P1 is mechanically connected to the second planet gear P2. There is no sun gear in the second layer. The first layer consisting of S, P1, and R1 needs sun gear because S connects to the input

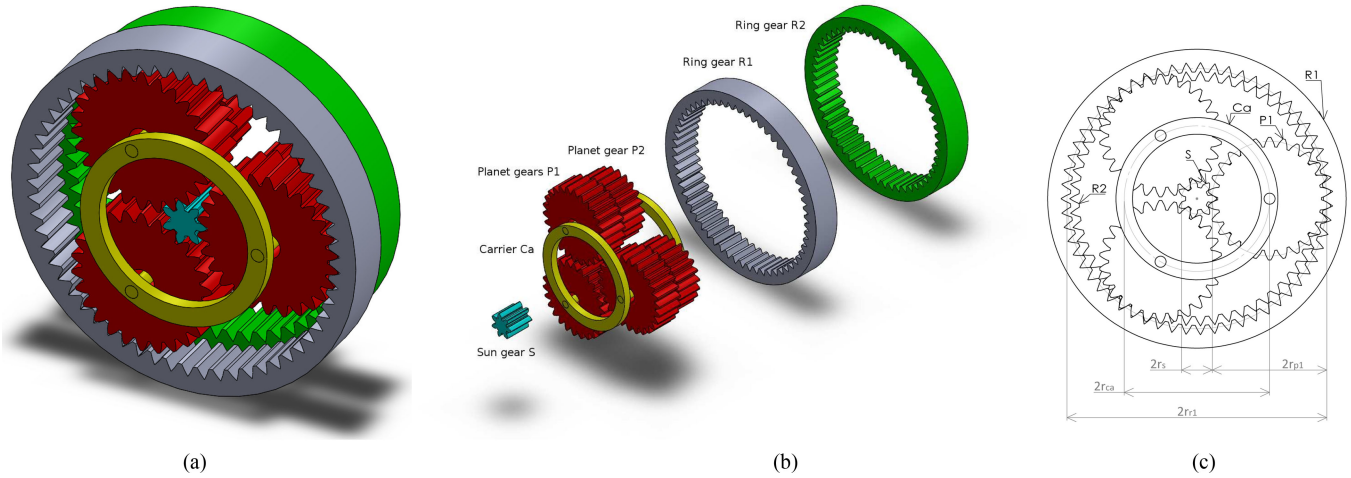


Fig. 3. Illustrative model of the type-3K compound planetary gearbox. (a) Assembled model. (b) Disassembled model. (c) Kinematics of gears.

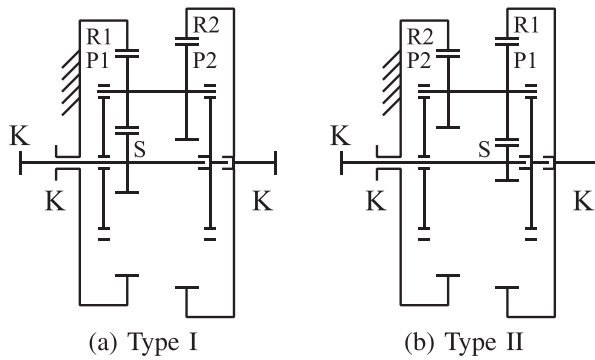


Fig. 4. Stick diagram of type-3K compound planetary gearbox. (a) Type I. (b) Type II.

shaft. However, the second layer that includes P2 and R2 does not have a sun gear. The planet gears in the first layer are directly connected to the planet gears in the second layer.

In the Type-I configuration [see Fig. 4(a)], the first sun gear is at the input side, the first ring gear R1 is fixed, and the second ring gear R2 is at the output side. In the Type-II configuration [see Fig. 4(b)], R2 is fixed and R1 is at the output side. A range of the reduction ratio can be enlarged by changing the number of teeth. However, this gearbox is not commonly used at high ratios because increasing the gear ratio reduces the driving efficiency. The basic driving efficiency and the gear ratio in the Type-I configuration are calculated as follows.

A. Kinematics

The velocity relationship between S and P1 on the pitch circle is

$$r_s \omega_s = (r_s + r_{p11}) \omega_{ca} - r_{p11} \omega_{p1} \quad (40)$$

where r_* is the radius of the pitch circle [or the carrier see Fig. 3(c)]. Note that r_{p11} and r_{p12} are the radius of the pitch circle of P1 when engaging to S and R1, respectively. In general, the

TABLE II
NOMENCLATURE OF RADII

r_{ca}	The radius of Carrier
r_s	The pitch circle of S
r_{r1}	The pitch circle of R1
r_{r2}	The pitch circle of R2
r_{p2}	The pitch circle of P2
r_{11}	The pitch circle of P1 when engaging to S
r_{12}	The pitch circle of P1 when engaging to R1
r_a	Center distance between S and P1
r_b	Center distance between P1 and R1
r_c	Center distance between P2 and R2

planetary gearbox allows different radii of the pitch circles for engaging S and R1. We have summarized all the radii in Table II. ω_{p1} , ω_{p2} , ω_{r1} , ω_{r2} , ω_s , and ω_{ca} are the counterclockwise angular velocities of the gear P1, P2, R1, R2, S, and carrier, respectively. Accordingly, the velocity relationships between P1 and R1 and between P2 and R2 on the pitch circle are, respectively, given by

$$r_{r1} \omega_{r1} = (r_{r1} - r_{p11}) \omega_{ca} + r_{p11} \omega_{p1} \quad (41)$$

$$r_{r2} \omega_{r2} = (r_{r2} - r_{p12}) \omega_{ca} + r_{p12} \omega_{p2}. \quad (42)$$

Moreover, the angular velocities of P1 and P2 are identical ($\omega_p = \omega_{p1} = \omega_{p2}$) because these gears are connected. The center-to-center distances of all gear pairs are also identical

$$r_s + r_{p11} = r_{r1} - r_{p12} = r_{r2} - r_{p2} = r_{ca}. \quad (43)$$

When R1 is fixed ($\omega_{r1} = 0$), the output angular velocity ω_{r2} is given by

$$\omega_{r2} = \frac{1 - I_2}{1 + I_1} \omega_s = G_{r2} \omega_s \quad (44)$$

$$I_1 = \frac{r_{r1} r_{p11}}{r_s r_{p12}} = \frac{z_{r1}}{z_s} \quad (45)$$

$$I_2 = \frac{r_{r1} r_{p2}}{r_{r2} r_{p12}} = \frac{z_{r1} z_{p2}}{z_{r2} z_{p1}}. \quad (46)$$

Here, the pitch-diameter ratio of a pair of gears equals the teeth-number ratio of the pair. The gear ratio is $G_{r2} = (1 - I_2)/(1 + I_1)$, where z_* is the number of teeth. This expression realizes a high gear ratio ($|G_{r2}| \ll 1$) because I_2 is nearly 1. By considering (44), when $I_2 < 1$, the rotation direction of the output shaft is the same as the rotation direction of the input shaft. On the other hand, when $I_2 > 1$, the rotation direction of the output shaft is inverse of the rotation direction of the input shaft.

B. Driving Efficiency

1) Driving Efficiency and Tangential Force of a Pair of Gears:

To derive the driving efficiency, one must consider the force loss between the paired gears. As the ratio of the pitch-circle velocities of a pair of involute gears is constant, the force is lost in the pitch-circle direction.

The basic driving efficiency η_0 of a pair of gears is calculated from the velocity relationship on the pitch circle and the relationship between the tangential force on the pitch circle and the torque τ_{in} . The velocity relationship is given by

$$r_{in}\omega_{in} = -r_{out}\omega_{out}. \quad (47)$$

The tangential force on the pitch circle is related to the torque as follows:

$$\tau_{in} = f_{in}r_{in} \quad (48)$$

$$\tau_{out} = -f_{out}r_{out}. \quad (49)$$

Here, r_{in} and r_{out} are the pitch circle radii of a driving and driven gear, respectively. ω_{in} and ω_{out} are the angular velocity of an active driving and a passive driven gear, respectively. The tangential force from the driving gear is f_{in} , and the tangential force received by the driven gear is f_{out} .

From (47)–(49), the basic driving efficiency η_0 is given by

$$\eta_0 = \frac{\omega_{out}\tau_{out}}{\omega_{in}\tau_{in}} = \frac{f_{out}}{f_{in}}. \quad (50)$$

When the gear is driving, f_{out} is given by $f_{out} = \eta_0 f_{in}$. The loss of force, $f_{in} - f_{out} = (1 - \eta_0)f_{in}$, corresponds to the friction force.

2) Forward-Driving Efficiency: The type-3K compound planetary gearbox has three meshing points: one between S and P1, the second between P1 and R1, and the third between P2 and R2. This section computes the efficiency of a gearbox in terms of the basic driving efficiencies of its gears. To this end, we derive the torque and power balance equations. This model extends (40)–(42) to express an imaginary slip on the pitch circle as follows:

$$v_{sp1} = r_s\omega_s - (r_s + r_{p11})\omega_{ca} + r_{p11}\omega_p \quad (51)$$

$$v_{r1p1} = r_{r1}\omega_{r1} - (r_{r1} - r_{p12})\omega_{ca} - r_{p12}\omega_p \quad (52)$$

$$v_{r2p2} = r_{r2}\omega_{r2} - (r_{r2} - r_{p2})\omega_{ca} - r_{p2}\omega_p \quad (53)$$

where v_{ij} is the tangential velocity of gear i relative to gear j . In matrix form, these equations are expressed as $\mathbf{v} = \mathbf{U}\boldsymbol{\omega}$, where

$\mathbf{v} = [v_{sp1} \ v_{r1p1} \ v_{r2p2}]^T$, $\boldsymbol{\omega} = [\omega_s \ \omega_p \ \omega_{r1} \ \omega_{r2} \ \omega_{ca}]^T$, and

$$\mathbf{U} = \begin{bmatrix} r_s & r_{p11} & 0 & 0 & -r_s - r_{p11} \\ 0 & -r_{p12} & r_{r1} & 0 & -r_{r1} + r_{p12} \\ 0 & -r_{p2} & 0 & r_{r2} & -r_{r2} + r_{p2} \end{bmatrix}. \quad (54)$$

If no energy is lost between each pair of gears, we obtain

$$\begin{bmatrix} \tau_s \\ \tau_p \\ \tau_{r1} \\ \tau_{r2} \\ \tau_{ca} \end{bmatrix} = \begin{bmatrix} r_s & 0 & 0 \\ r_{p11} & -r_{p12} & -r_{p2} \\ 0 & r_{r1} & 0 \\ 0 & 0 & r_{r2} \\ -r_s - r_{p11} & -r_{r1} + r_{p12} & -r_{r2} + r_{p2} \end{bmatrix} \begin{bmatrix} f_{sp1} \\ f_{r1p1} \\ f_{r2p2} \end{bmatrix}. \quad (55)$$

By setting variables $\boldsymbol{\tau} = [\tau_s \ \tau_p \ \tau_{r1} \ \tau_{r2} \ \tau_{ca}]^T$ and $\mathbf{f} = [f_{sp1} \ f_{r1p1} \ f_{r2p2}]^T$, (55) is expressed as $\boldsymbol{\tau} = \mathbf{U}^T \mathbf{f}$. τ_i is the torque of gear i , and f_{ij} is the tangential force on the pitch circle exerted by gear j on gear i . When the kinematic relationship is $\mathbf{v} = 0$, the torque balance equation is $\boldsymbol{\tau} = 0$. This result is easily confirmed by the virtual work principle $\boldsymbol{\tau}^T \boldsymbol{\omega} = \mathbf{f}^T \mathbf{v}$. As no external torque is supplied to P1, P2, and Ca (carrier), the forces between a pair of gears f_{sp1} , f_{r1p1} , and f_{r2p2} are derived from (55) with $\tau_p = \tau_{ca} = 0$ as follows:

$$f_{sp1} = \frac{1}{r_s} \tau_s \quad (56)$$

$$f_{r1p1} = \frac{I_1 + I_2}{r_{r1}(1 - I_2)} \tau_s \quad (57)$$

$$f_{r2p2} = -\frac{1 + I_1}{r_{r2}(1 - I_2)} \tau_s. \quad (58)$$

These equations show the direction of the power transfer between a planet gear and a ring gear, considering the value of I_2 .

To discuss the basic driving efficiency, we next consider the relative motions of a pair of gears from the carrier perspective. From (40)–(42), we obtain $\omega_{ca} = \omega_s/(1 + I_1)$. Then, the relative velocities of S, R1, and R2 with respect to the carrier are, respectively, given by

$$\omega_s - \omega_{ca} = \frac{I_1}{1 + I_1} \omega_s \quad (59)$$

$$\omega_{r1} - \omega_{ca} = -\frac{1}{1 + I_1} \omega_s \quad (60)$$

$$\omega_{r2} - \omega_{ca} = -\frac{I_2}{1 + I_1} \omega_s. \quad (61)$$

Based on these results, the relative power flows as follows:

- 1) The relative power is always transferred from S to P1 because the signs of (56) and (59) are the same.
- 2) If $I_2 < 1$, the relative power is transferred from P1 to R1 because the signs of (57) and (60) are opposite. If $I_2 > 1$, the power is transferred from R1 to P1 because the signs of (56) and (59) are the same.
- 3) If $I_2 < 1$, the relative power is transferred from R2 to P2 because the signs of (58) and (61) are the same. If $I_2 > 1$,

the power is transferred from P2 to R2 because the signs of (58) and (61) are opposite.

Note that R1 is stationary.

Based on the basic driving efficiency in Section III-B and the direction of the power flow above, we now correct (55) to account for loss of power and friction between the driving and driven gears. The driven force decreases in proportion to the basic driving efficiency, as shown in (50). For example, when tangential force exerted by the driving gear S to the driven gear P1 is f_{sp1} , the reactive tangential force received by P1 becomes $\eta_a f_{sp1}$, where η_a is the basic efficiency between S and P1. This means that Newton's third law of motion considers friction $(1 - \eta_a)f_{sp1}$.

Considering the basic driving efficiencies between S and P1, R1 and P1, and R2 and P2, the balanced torque equation (55) is modified as follows. Note that $z_i, i \in \{p1, p2, r1, r2, s\}$, is the number of teeth.

a) Case of $I_2 < 1$ ($z_{r1} > z_{r2}$ and $z_{p1} > z_{p2}$): Because the relative power propagates from S to P1, from P1 to R1, and from R2 to P2, the torques are given by

$$\begin{bmatrix} \tau_s \\ \tau_p \\ \tau_{r1} \\ \tau_{r2} \\ \tau_{ca} \end{bmatrix} = \begin{bmatrix} r_s & 0 & 0 \\ r_{p11}\eta_a & -r_{p11} & -r_{p2}\eta_c \\ 0 & r_{r1}\eta_b & 0 \\ 0 & 0 & r_{r2} \\ -r_s - r_{p11}\eta_a & -r_{r1}\eta_b + r_{p12} & -r_{r2} + r_{p2}\eta_c \end{bmatrix} \times \begin{bmatrix} f_{sp1} \\ f_{r1p1} \\ f_{r2p2} \end{bmatrix}. \quad (62)$$

Note that η_a , η_b , and η_c are basic driving efficiencies between S and P1, between R1 and P1, and between R2 and P2, respectively. Eliminating f_{sp1} , f_{r1p1} , f_{r2p2} , and τ_{r1} from the above equations and assuming no propagation of power from outside the planet gear and the carrier ($\tau_p = \tau_{ca} = 0$), the torque τ_{r2} of R2 is obtained as

$$\tau_{r2} = -\frac{1 + \eta_a\eta_b I_1}{1 - \eta_b\eta_c I_2} \tau_s. \quad (63)$$

Setting τ_s as the input torque τ_{in} , the output τ_{out} is equal to $-\tau_{r2}$. The forward-driving efficiency η of the gearbox is

$$\eta = \frac{\tau_{out}\omega_{r2}}{\tau_{in}\omega_s} = \frac{(1 + \eta_a\eta_b I_1)(1 - I_2)}{(1 + I_1)(1 - \eta_b\eta_c I_2)} \quad (64)$$

where we have used (44) and (63).

Fig. 5 shows free-body diagrams of the proposed 3K gearbox in the case of forward driving with $I_2 < 1$. Fig. 5(a) shows the rotation of the gears and the carrier in the stationary frame. The sun gear S, the carrier, and the ring gear R2 rotate counterclockwise, while the planet gears P1 and P2 rotate clockwise according to (40)–(42). When the external torque is supplied to S to drive it counterclockwise and there is no energy loss, the tangential force on the pitch circle exerted on each gear is shown in the figure. The directions of force are determined by (56)–(58). Fig. 5(b) shows the rotation of the gears in the rotating reference frame fixed on the carrier. The direction of the force

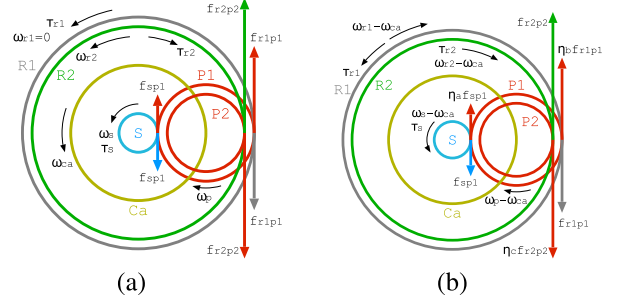


Fig. 5. Free-body diagram of the proposed 3K gearbox in a case of forward driving with $I_2 < 1$. (a) Rotation of the gears and the carrier in the stationary frame and the tangential force on the pitch circle when there is no energy loss. (b) Rotation of the gears in the rotating reference frame fixed on the carrier and the tangential force on the pitch circle when there is energy loss.

is same in Fig. 5(a), but the magnitude is slightly decreased due to the energy loss according to the discussion of the flow of the relative power. In terms of the planet gear, the equilibrium condition $\tau_p = r_{p11}f_{sp1} - r_{p12}f_{r1p1} - r_{p2}f_{r2p2} = 0$ holds, where $f_{sp1} > 0$, $f_{r1p1} > 0$, and $f_{r2p2} < 0$ when $I_2 < 1$. The “relative power” transfers from P1 to R1 and R2 to P2 in the rotating reference frame fixed on the carrier. These relative power flows are only used in the analysis for the basic efficiency between a pair of gears. In the stationary frame, of course, no power flows between P1 and R1, and the real power transfers from P2 to R2.

b) Case of $I_2 > 1$ ($z_{r1} < z_{r2}$ and $z_{p1} < z_{p2}$): Because the relative power propagates from S to P1, from R1 to P1, and from P2 to R2, in the same manner the case of $I_2 < 1$, the torques are given by

$$\begin{bmatrix} \tau_s \\ \tau_p \\ \tau_{r1} \\ \tau_{r2} \\ \tau_{ca} \end{bmatrix} = \begin{bmatrix} r_s & 0 & 0 \\ r_{p11}\eta_a & -r_{p12}\eta_b & -r_{p2} \\ 0 & r_{r1} & 0 \\ 0 & 0 & r_{r2}\eta_c \\ -r_s - r_{p11}\eta_a & -r_{r1} + r_{p12}\eta_b & -r_{r2}\eta_c + r_{p2} \end{bmatrix} \times \begin{bmatrix} f_{sp1} \\ f_{r1p1} \\ f_{r2p2} \end{bmatrix}. \quad (65)$$

The torque τ_2 of the ring gear R2 is

$$\tau_{r2} = -\frac{\eta_c(\eta_b + \eta_a I_1)}{\eta_b\eta_c - I_2} \tau_s. \quad (66)$$

From the above equations, the forward-driving efficiency η is obtained as

$$\eta = \frac{\eta_c(\eta_b + \eta_a I_1)(1 - I_2)}{(1 + I_1)(\eta_b\eta_c - I_2)}. \quad (67)$$

3) Backward-Driving Efficiency: The backward-driving efficiency is calculated identically to the forward-driving efficiency. From (55) and the torque τ_{r2} , the works done on R2 by the forces between the gear pair f_{sp1} , f_{r1p1} , and f_{r2p2} are

given by

$$f_{sp1} = -\frac{I_1(1 - I_2)}{r_{r1}(1 + I_1)}\tau_{r2} \quad (68)$$

$$f_{r1p1} = -\frac{I_1 + I_2}{r_{r1}(1 + I_1)}\tau_{r2} \quad (69)$$

$$f_{r2p2} = \frac{1}{r_{r2}}\tau_{r2}. \quad (70)$$

Furthermore, the velocities relative to the carriers of S, R1, and R2 are, respectively, given by

$$\omega_s - \omega_{ca} = \frac{I_1}{1 - I_2}\omega_{r2} \quad (71)$$

$$\omega_{r1} - \omega_{ca} = -\frac{1}{1 - I_2}\omega_{r2} \quad (72)$$

$$\omega_{r2} - \omega_{ca} = -\frac{I_2}{1 - I_2}\omega_{r2}. \quad (73)$$

According to the above equations, the direction of relative power flows are as follows.

- 1) The relative power is always transferred from P1 to S, because the signs of (68) and (71) are opposite.
- 2) if $I_2 < 1$, the power is transferred from R1 to P1, because the signs of (69) and (72) are the same. If $I_2 > 1$, the power is transferred from P1 to R1, because the signs of (69) and (72) are opposite.
- 3) if $I_2 < 1$, the power is transferred from P2 to R2, because the signs of (70) and (73) are opposite. If $I_2 > 1$, the power is transferred from R2 to P2, because the signs of (70) and (73) are the same.

a) Case of $I_2 < 1$ ($z_{r1} > z_{r2}$ and $z_{p1} > z_{p2}$): Because the relative power propagates from P1 to S, from R1 to P1, and from P2 to R2, the torque balance equations become

$$\begin{bmatrix} \tau_s \\ \tau_p \\ \tau_{r1} \\ \tau_{r2} \\ \tau_{ca} \end{bmatrix} = \begin{bmatrix} r_s\eta_a & 0 & 0 \\ r_{p11} & -r_{p12}\eta_b & -r_{p2} \\ 0 & r_{r1} & 0 \\ 0 & 0 & r_{r2}\eta_c \\ -r_s\eta_a - r_{p11} & -r_{r1} + r_{p12}\eta_b & -r_{r2}\eta_c + r_{p2} \end{bmatrix} \times \begin{bmatrix} f_{sp1} \\ f_{r1p1} \\ f_{r2p2} \end{bmatrix}. \quad (74)$$

From the above equations, the backward-driving efficiency η' of the gearbox is obtained as

$$\eta' = \frac{\tau_{out}\omega_s}{\tau_{in}\omega_{r2}} = \frac{(1 + I_1)\eta_a(\eta_b\eta_c - I_2)}{\eta_c(\eta_a\eta_b + I_1)(1 - I_2)} \quad (75)$$

where the input torque $\tau_{in} = \tau_{r2}$, and the output torque $\tau_{out} = -\tau_s$. Note that (75) can calculate in the same manner of the forward-driving efficiency. When $\eta_b\eta_c \leq I_2$, the backward-driving efficiency η' is below 0, and the gearbox cannot be backdriven. Thus, the condition under which the gearbox can perform backdrive is $\eta_b\eta_c > I_2$.

b) Case of $I_2 > 1$ ($z_{r1} < z_{r2}$ and $z_{p1} < z_{p2}$): The relative power is transferred from P1 to S, from P1 to R1, and from R2

to P2. When $I_2 > 1$, the backward-driving efficiency η' of the gearbox is given by

$$\eta' = \frac{\tau_{out}\omega_s}{\tau_{in}\omega_{r2}} = \frac{(1 + I_1)\eta_a(1 - \eta_b\eta_c I_2)}{(\eta_a + \eta_b I_1)(1 - I_2)}. \quad (76)$$

The condition under which the gearbox can perform backdrives is $\eta_b\eta_c > 1/I_2$. When R2 is fixed [the Type II configuration in Fig. 4(b)], it can be solved by the same procedure.

4) *Basic Driving Efficiency Between a Pair of Gears*: The driving efficiencies of the gearbox are a function of the teeth ratios I_1 and I_2 and the basic driving efficiencies of gear pairs. This subsection describes the basic driving efficiencies. The basic driving efficiencies between S and P1, between R1 and P1, and between R2 and P2, denoted by η_a , η_b , and η_c , respectively, are given by [37]

$$\eta_i = 1 - \mu\pi \left(\frac{1}{z_{i1}} + \text{sgn}_i \frac{1}{z_{i2}} \right) \epsilon_i, \quad i \in \{a, b, c\} \quad (77)$$

where the subscripts are defined by $a1 = s$, $a2 = p1$, $b1 = p1$, $b2 = r1$, $c1 = p2$, and $c2 = r2$, and sign parameters are given by $\text{sgn}_a = 1$ and $\text{sgn}_b = \text{sgn}_c = -1$. These definitions are also used in the following equations in this section. In addition, μ is the average friction coefficient of the teeth surface, and the parameters ϵ_i for $i \in \{a, b, c\}$ are given by

$$\epsilon_i = \epsilon_{i1}^2 + \epsilon_{i2}^2 - \epsilon_{i1} - \epsilon_{i2} + 1 \quad (78)$$

$$\epsilon_{i1} = \text{sgn}_i \frac{z_{i2}}{2\pi} (\tan \alpha_{ai2} - \tan \alpha_{wi}) \quad (79)$$

$$\epsilon_{i2} = \frac{z_{i1}}{2\pi} (\tan \alpha_{ai1} - \tan \alpha_{wi}) \quad (80)$$

for $i \in \{a, b, c\}$, where ϵ_{i1} and ϵ_{i2} are the approach and recess contact ratio, respectively. α_{wi} is the working pressure angle, and α_{ai1} and α_{ai2} are the tip pressure angles.

Fig. 6 shows the geometric meaning of each parameter [38]. The working pressure angle is obtained for $i \in \{a, b, c\}$ as

$$\cos \alpha_{wi} = \frac{d_{bi2} + \text{sgn}_i d_{bi1}}{2r_i} = \frac{m_i(z_{i1} + \text{sgn}_i z_{i2}) \cos \alpha}{2r_i} \quad (81)$$

where r_a , r_b , and r_c are the center distances between S and P1, between P1 and R1, and between P2 and R2, respectively (see Fig. 6). During the optimization process, a constraint for the center distance $r_a = r_b = r_c$ is introduced. For $i \in \{a, b, c\}$, $d_{bi1} = m_i z_{i1} \cos \alpha$ and $d_{bi2} = m_i z_{i2} \cos \alpha$ are the basic circle diameters, m_i is the module of the corresponding meshing gears, and α is the basic pressure angle. Note that $m_a = m_b$. The tip pressure angle is given by $\cos \alpha_{aj} = \frac{d_{bj}}{d_{aj}}$, $j \in \{s, p1, r1, p2, r2\}$, where d_{aj} is the tip circle diameter.

The tip circle diameters of each gear are given by

$$d_{as} = m_a z_s + 2m_a(1 + y_a - x_{p1}) \quad (82)$$

$$d_{ap1} = m_a z_{p1} + 2m_a(1 + \min(y_a - x_s, x_{p1})) \quad (83)$$

$$d_{ar1} = m_a z_{r1} - 2m_a(1 - x_{r1}) \quad (84)$$

$$d_{ap2} = m_c z_{p2} + 2m_c(1 + x_{p2}) \quad (85)$$

$$d_{ar2} = m_c z_{r2} - 2m_c(1 - x_{r2}) \quad (86)$$

where x_j , $j \in \{s, p1, r1, p2, r2\}$, is the profile shift coefficient, and y_a is the center distance modification coefficient. This

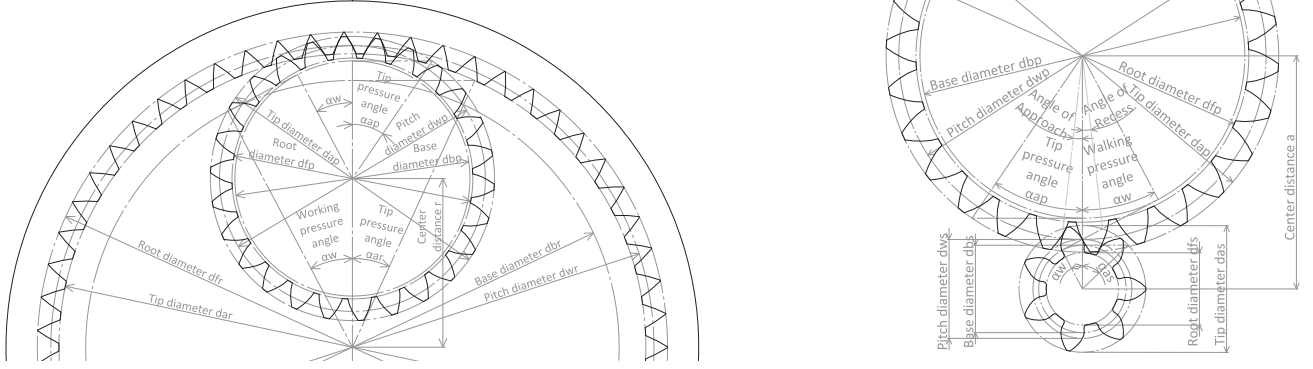


Fig. 6. Parameters of the paired gears.

coefficient is calculated as

$$y_a = \frac{z_s + z_{p1}}{2} \left(\frac{\cos \alpha}{\cos \alpha_{wa}} - 1 \right). \quad (87)$$

The profile shift coefficients of each gear are described by the following relationships:

$$x_{i1} + \text{sgn}_i x_{i2} = \frac{z_{i1} + \text{sgn}_i z_{i2}}{2 \tan \alpha} (\text{inv} \alpha_{wi} - \text{inv} \alpha) \quad (88)$$

where $\text{inv} \alpha = \tan \alpha - \alpha$ is the involute function. In summary, the forward- and backward-driving efficiencies of the gear system depend on the number of teeth, the profile shift coefficients, and the basic pressure angles. From (81) and (88), the forward- and backward-driving efficiencies are decided by z_{p1} , z_{r1} , z_{p2} , z_{r2} , x_{r1} , x_{r2} , and α . In most cases, α is 20° .

IV. MAXIMIZATION OF TOTAL EFFICIENCY

The optimization problem of forward-driving efficiency is formulated by (89) and (91). The decision variables are the number of teeth and the profile shift coefficients

$$\text{maximize } \eta(z_s, z_{p1}, z_{r1}, z_{p2}, z_{r2}, x_s, x_{p1}, x_{r1}, x_{p2}, x_{r2}) \quad (89)$$

$$\text{subject to } G_L \leq G_{r2} \leq G_U \quad (90)$$

$$-2 \leq x_i \leq 2, i \in \{s, p1, r1, p2, r2\} \quad (91)$$

$$r_a = r_b = r_c \quad (92)$$

where G_{r2} is the gear ratio defined by (44) as a function of the numbers of teeth, G_L and G_U are lower and upper bounds of the target gear ratio, and $z_s, z_{p1}, z_{r1}, z_{p2}, z_{r2} \in \mathbb{Z}$, $x_s, x_{p1}, x_{r1}, x_{p2}, x_{r2} \in \mathbb{R}$. In addition, the conditions for Trochoid interference should be satisfied. The target function of η is (64) or (67). Equations (91) and (92) give the general constraints on the profile shift and center distance, respectively. When producing the gearbox, we must also consider the assembly conditions and the distances between the planet gears. We must also consider the conditions for the contact ratios. The contact ratio between S and P2 tends to be less than 1.0 when we discard these conditions. The contact ratios between P1 and R1 and

between P2 and R1 are usually larger than 1.0 regardless of these conditions. When the tooth number is fixed, the maximization problem becomes convex, as shown in Fig. 7. Fig. 7(d) shows the efficiency versus one of the profile shift coefficients. By optimization, the driving efficiency increases by around 15%. The dependent variable in the optimization function is the forward-driving efficiency (z -axis), and independent variables are x_{r1} , x_{r2} , r_a , r_b , and r_c . The above optimization problem was solved by the quasi-Newton method [39]–[46]. When the forward-driving efficiency was maximized, the backward-driving efficiency was also maximized.

When we decide the specifications of the gearbox and conduct the optimization process, the following steps have been taken.

- 1) Set the lower and upper bounds of the reduction ratio of the target gear.
- 2) Set the minimum number of teeth for S and the maximum number of teeth for R1 and R2.
- 3) Apply exhaustive search for the numbers of teeth, z_s , z_{p1} , z_{r1} , z_{p2} , and z_{r2} .
 - a) Check the feasibility of the temporary solution for the conditions for the gear ratio and the center distance in the exhaustive search.
 - b) If it is feasible, optimize the efficiency w.r.t. continuous variables using the quasi-Newton method in every feasible solution in the exhaustive search. In practice, x_s , x_{p1} , and x_{p2} are explicitly described by x_{r1} , x_{r2} , and r_c according to (81), (88), and (92).
 - c) Check if all the contact ratios are between 1.0 and 2.0.
 - d) Calculate the allowable torque of the gearbox based on the tooth strength defined by the standard of Japan Gear Manufacturers Association and check if it satisfies the requirement.
- 4) Find the best solution among the feasible solutions.

The maximization of the efficiency is equivalent to the minimization of the frictional power loss due to the meshing of the gearing components. When we limit the upper bound of the profile shift coefficient up to 0.9, the tooth shape becomes reasonable, and the optimized efficiency degrades around 1% compared to the case of the upper bound of 2.0.

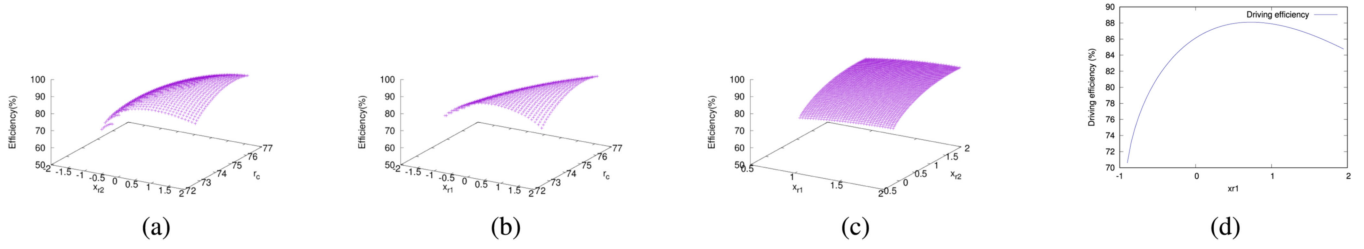


Fig. 7. Solution of the maximization problem for fixed tooth number. (a)–(c) Shows the shape of the function of the driving efficiency w.r.t. decision variables. The z -axis plots the driving efficiency when (a) x_{r1} is fixed, (b) x_{r2} is fixed, and (c) r_c is fixed. (d) Driving efficiency versus x_{r1} when x_{r2} and r_c are fixed.

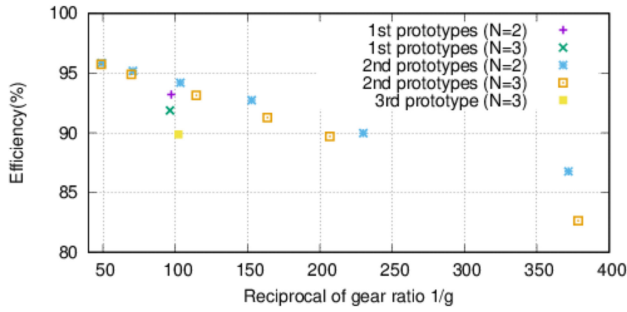


Fig. 8. Expected efficiency profiles in prototypes with N planet gears.

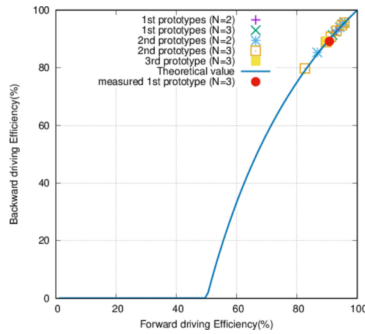


Fig. 9. Backward versus forward efficiencies of the prototypes (colored symbols), the theoretical curve (solid cyan line), and the measured efficiency of the first prototype with a gear ratio of $1/96.2$ ($N = 3$) (red symbol).

A. Performance of the Prototype Gearbox

The proposed optimization algorithm was tested in a prototype gearbox called Bilateral Drive Gear. We made first prototypes ($N = 2$ and 3) at first, second prototypes ($N = 2$ and 3) second, and third prototypes ($N = 3$) at last in order, which have different gear ratios and the different numbers of planet gears. The designed efficiency profiles of prototypes with N planet gears (in this case, two or three) are presented in Fig. 8. Fig. 9 plots the designed backward- versus forward-driving efficiencies and the theoretical efficiency curve [see Fig. 2(a)] for comparison. All prototypes achieved high-efficiency operation. The expected efficiencies exceed 90% even at a high gear ratio ($>1/200$).

TABLE III
SPECIFICATIONS OF THE FIRST PROTOTYPE WITH A GEAR RATIO OF $1/96.2$ ($N = 3$)

Parameter	value
z_s	12
z_{p1}	39
z_{r1}	90
z_{p2}	32
z_{r2}	81
N	3
m_a [mm]	0.8
m_c [mm]	0.8467
α [deg]	20
x_s	0.476
x_{p1}	0.762
x_{p2}	0.536
x_{r1}	2.000
x_{r2}	1.210
contact ratio between S and P1	1.232
contact ratio between P1 and R1	1.420
contact ratio between P2 and R2	1.565
Materials of gears and shaft	SCM440
Materials of a case and carrier	A5052

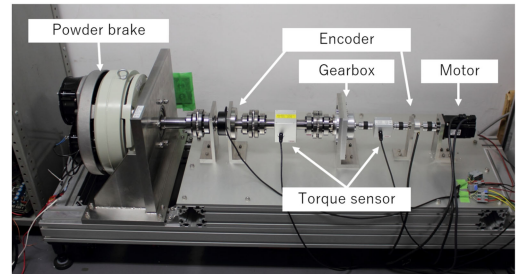


Fig. 10. Experiment environment for measuring the driving efficiency. The right side is the input side of the gearbox.

We measured the specifications of the first prototype with the gear ratio $1/96.2$ ($N = 3$). This gearbox is the first prototype ($N = 3$). The measured driving efficiency is 89.0%; the estimated basic efficiency are $\eta_a = 97.7\%$, $\eta_b = 99.6\%$, and $\eta_c = 99.7\%$. Table III gives the specifications of the first prototype gearbox with a gear ratio of $1/96.2$ ($N = 3$).

The experiment environment is illustrated in Fig. 10 for measuring the driving efficiency. We used a motor (SGMJV-04 A YASKAWA), an encoder (MTL MG30), and a torque sensor (UTMII-2Nm) in the input side of the gearbox. In the output side of the gearbox, we used a load (ZKB-20HBN MITSUBISHI), an

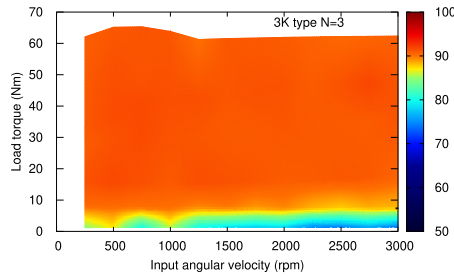


Fig. 11. Measured forward-driving efficiency map of the first prototype with a gear ratio of 1/96.2 ($N = 3$). The color bar shows the efficiency (%).

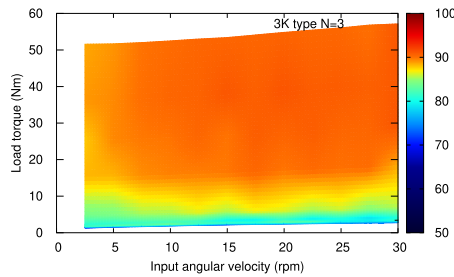


Fig. 12. Measured backward-driving efficiency map of the first prototype with a gear ratio of 1/96.2 ($N = 3$). The color bar shows the efficiency (%).

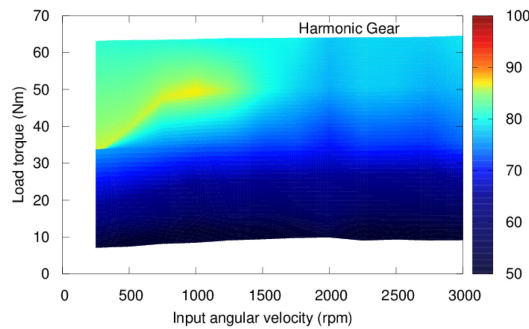


Fig. 13. Measured forward-driving efficiency map of the harmonic gear. The color bar shows the efficiency (%).

encoder (MTL MEH-60), and a torque sensor (UTMII-200Nm). The motor is operated using velocity control. We used PR-Expert4 [47] for digital control. In these experiments, the control cycle is 100 μ s. The encoder resolution of input and output sides are 300 and 108 000 P/R, respectively, when measuring the efficiency. When measuring backlash, we used two encoders with an encoder resolution of 11 840 000 P/R at both sides. In these experiments, we applied the grease with NLGI no.2 in this gearbox.

The measured forward- and backward-driving efficiencies of the prototype and forward-driving efficiency of the harmonic gearbox are plotted as functions of the input angular velocity in Figs. 11–13, respectively. The average of the forward- and backward-driving efficiencies was 89.0% and 85.3%, respectively. The reverse-drive starting torque was 0.020 N·m. The driving efficiency of the same gearbox with uncorrected teeth is

TABLE IV

PERFORMANCE COMPARISON RESULTS BETWEEN THE FIRST PROTOTYPE WITH A GEAR RATIO OF 1/96.2 ($N = 3$) AND HARMONIC GEARBOXES

	Proposed gearbox	Harmonic gearbox CSG25 lightweight unit type
Forward driving efficiency (%)	89.0	67.9
Backward driving efficiency (%)	85.3	–
Gear ratio	1/96.2	1/100
Allowable torque (N.m)	120	87
Reverse drive starting torque (N.m)	0.020	11
Backlash (arc-min)	24	0.15
Diameter (mm)	94	86
Height (mm)	62	46
Weight (kg)	1.33	1.1

68.5% (calculated value). It is 20.5% lower than the optimized one.

Table IV compares the performances of the prototype gearbox and a harmonic gearbox (CSG 17 lightweight unit type [34]). The harmonic gearbox is used in diverse robots, such as industrial robots, amusement robots, peripheral robot equipment, and humanoid robots [48], [49].

The forward-driving efficiency was 21.1% higher in our prototype than in the harmonic gearbox with a similar gear ratio. The reverse-drive starting torque in the prototype gearbox was quite small. These results confirm good backdrivability of the prototype gearbox. This gearbox is easily rotated backward by hand.

As the benefits of this bilateral drive gear, sensorless torque control for a powered exoskeleton has been proposed [50]. In this article, the bilateral drive gear with a reduction ratio of 1/102.1 has been used. By using the multienncoder-based disturbance observer, the system can estimate the torque of the knee of the exoskeleton without torque sensor.

V. SUMMARY

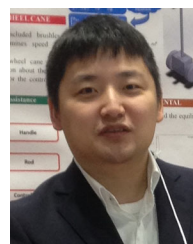
We maximized the forward- and backward-driving efficiencies of a type-3K compound planetary gearbox and tested the performances of a prototype gearbox.

The prototype gearbox achieved 89.0% forward-driving efficiency, 85.3% backward-driving efficiency, and a 0.020-N·m reverse-drive starting torque. We also developed prototype gearboxes with gear ratios between 1/48.7 and 1/378.5. All of these prototypes were manually backdrivable. Our bilateral drive gears are expected to be employed in cooperative robots. However, increasing gear ratio increases the apparent inertia [22]. As the future work, we will investigate the effects of this apparent inertia. In future, we plan to develop the actuator with this bilateral drive gear for cooperative robots.

REFERENCES

- [1] S. R. Gomez, V. M. Becerra, J. R. Llata, E. González-Sarabia, C. Torre-Ferrero and J. Pérez-Oria, "Working together: A review on safe human-robot collaboration in industrial environments," *IEEE Access*, vol. 5, pp. 26754–26773, 2017.
- [2] L. Adrian *et al.*, "Robotic green asparagus selective harvesting," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 6, pp. 2401–2410, Dec. 2017.

- [3] H. Siebald *et al.*, "Real-time acoustic monitoring of cutting blade sharpness in agricultural machinery," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 6, pp. 2411–2419, Dec. 2017.
- [4] E. Asadi, B. Li, and I. M. Chen, "Pictobot: A cooperative painting robot for interior finishing of industrial developments," *IEEE Robot. Autom. Mag.*, vol. 25, no. 2, pp. 82–94, Jun. 2018.
- [5] A. Cencen, J. C. Verlinden, and J. M. P. Geraedts, "Design methodology to improve human-robot coproduction in small- and medium-sized enterprises," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 3, pp. 1092–1102, Jun. 2018.
- [6] A. Parri *et al.*, "Real-time hybrid locomotion mode recognition for lower limb wearable robots," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 6, pp. 2480–2491, Dec. 2017.
- [7] C. Nabeshima, K. Ayusawa, C. Hochberg, and E. Yoshida, "Standard performance test of wearable robots for lumbar support," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2182–2189, Jul. 2018.
- [8] J. Choo and J. H. Park, "Increasing payload capacity of wearable robots using linear actuators," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 4, pp. 1663–1673, Aug. 2017.
- [9] J. Huang, X. Tu, and J. He, "Design and evaluation of the RUPERT wearable upper extremity exoskeleton robot for clinical and in-home therapies," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 46, no. 7, pp. 926–935, Jul. 2017.
- [10] S. Hauser, M. Robertson, A. Ijspeert, and J. Paik, "JammJoint: A variable stiffness device based on granular jamming for wearable joint support," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 849–855, Apr. 2017.
- [11] L. Randazzo, I. Iturrate, S. Perdikis and J. D. R. Millán, "mano: A wearable hand exoskeleton for activities of daily living and neurorehabilitation," *IEEE Robot. Autom. Lett.*, vol. 3, no. 1, pp. 500–507, Jan. 2018.
- [12] W. Huo, S. Mohammed, J. C. Moreno, and Y. Amirat, "Lower limb wearable robots for assistance and rehabilitation: A state of the art," *IEEE Syst. J.*, vol. 10, no. 3, pp. 1068–1081, Sep. 2016.
- [13] K. Hori, "Differential planet gear unit," U.S. Patent 4 942 781. 1990.
- [14] K. Hori, "Planetary gear having its orbital gears out of phase," U.S. Patent 4 864 893. 1989.
- [15] G. A. Pratt and M. M. Williamson, "Series elastic actuators," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. Human Robot Interact. Cooperative Robots*, 1995, vol. 1, pp. 399–406.
- [16] M. Zinn, O. Khatib, B. Roth, and J. K. Salisbury, "Playing it safe," *IEEE Robot. Autom. Mag.*, vol. 11, no. 2, pp. 12–21, Jun. 2004.
- [17] C. A. Ihrke *et al.*, "Rotary series elastic actuator," U.S. Patent US8 291 788B2, 2011.
- [18] M. Laffranchi, N. Tzarakakis, and D. G. Caldwell, "A compact compliant actuator (compact) with variable physical damping," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2011, pp. 4644–4650.
- [19] A. Albu-Schäffer, S. Haddadin, C. Ott, A. Stemmer, T. Wimböck, and G. Hirzinger, "The DLR lightweight robot—Design and control concepts for robots in human environments," *Ind. Robot*, vol. 34, no. 5, pp. 376–385, 2007.
- [20] T. Kawakami, K. Ayusawa, H. Kaminaga, and Y. Nakamura, "High-fidelity joint drive system by torque feedback control using high precision linear encoder," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2010, pp. 3904–3909.
- [21] Y. Kuroki, Y. Kosaka, T. Takahashi, E. Niwa, H. Kaminaga, and Y. Nakamura, "Cr-N alloy thin-film based torque sensors and joint torque servo systems for compliant robot control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2013, pp. 4954–4959.
- [22] P. M. Wensing, A. Wang, S. Seok, D. Otten, J. Lang, and S. Kim, "Proprioceptive actuator design in the MIT Cheetah: Impact mitigation and high-bandwidth physical interaction for dynamic legged robots," *IEEE Trans. Robot.*, vol. 33, no. 3, pp. 509–522, Jun. 2017.
- [23] G. Kenneally, A. De, and D. E. Koditschek, "Design principles for a family of direct-drive legged robots," *IEEE Robot. Autom. Lett.*, vol. 1, no. 2, pp. 900–907, Jul. 2016.
- [24] D. Kobuse and Y. Fujimoto, "Efficiency optimization of high-reduction-ratio planetary gears for very high power density actuators," in *Proc. IEEE 25th Int. Symp. Ind. Electron.*, 2016, pp. 1240–1245.
- [25] Y. Fujimoto and D. Kobuse, "Highly backdrivable robotic actuators," in *Proc. IEEE Int. Workshop Sens., Actuation, Motion Control, Optim.*, Nagaoka, Japan, Mar. 2017.
- [26] V. K. Manglik, *Elements of Mechanical Engineering*. Ludhiana, India: Katson Publishing House, 1985.
- [27] Q. An, S. Suo, J. Yang, C. Wu, and C. Yan, "Program design for the 3K planetary gear mechanism with more than three planet gears," *IOP Confe. Ser. Mater. Sci. Eng.*, vol. 394, 2018, Art. no. 032127.
- [28] F. Faulhaber, "A second look at Gearbox efficiencies," *Mach. Des.*, vol. 74, pp. 82–84, 2002.
- [29] K. Arnaudov and D. Karaivanov, "The Wolfrom Gear Train: A case of highest-complexity related modifications of the tooth meshing," in *Proc. ASME Int. Des. Eng. Tech. Conf. Comput. Inf. Eng. Conf.*, 2009, pp. 89–94.
- [30] *CSF Series Component Type*, Harmonic Drive Systems, Inc., Tokyo, Japan, Jul. 2018. [Online]. Available: https://www.hds.co.jp/english/products/lineup/hd/01sr01_csf_2a/index.html/
- [31] *General Catalog*, Harmonic Drive Systems, Inc., Tokyo, Japan, Jul. 2018. [Online]. Available: http://hds-tech.jp/pdf/hd01_en.pdf
- [32] A. Wang and S. Kim, "Directional efficiency in geared transmissions: Characterization of backdrivability towards improved proprioceptive control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2015, pp. 1055–1062.
- [33] K. Deb and S. Jain, "Multi-speed gearbox design using multi-objective evolutionary algorithms," *J. Mech. Des.*, vol. 125, pp. 609–619, 2003.
- [34] *Harmonic Drive General Catalog*, Harmonic Drive Systems, Inc., Tokyo, Japan, Aug. 2019. [Online]. Available: <http://www.hds.co.jp/>
- [35] *Cyclo Gearbox Product Catalog*, Sumitomo Drive Technologies, Inc., Chesapeake, VA, USA, Aug. 2019. [Online]. Available: <https://cyclo.shi.co.jp/document-download/>
- [36] *RV Series Product Catalog*, Nabtesco Inc., Tokyo, Japan, Aug. 2019. [Online]. Available: <https://precision.nabtesco.com/en/download/>
- [37] T. Yada, "Review of gear efficiency equation and force treatment," *JSME Int. J. Ser. C, Dyn., Control, Robot., Des. Manuf.*, vol. 40, no. 1, pp. 1–8, 1997.
- [38] R. C. Juvinall and K. M. Marshek, *Fundamentals of Machine Component Design*. New York, NY, USA: Wiley, 2005.
- [39] W. C. Davidon, "Variable metric method for minimization," Argonne Nat. Lab., Lemont, IL, USA, AEC Res. Develop. Rep. ANL-5990, 1959.
- [40] R. Fletcher and M. J. D. Powell, "A rapidly convergent descent method for minimization," *Comput. J.*, vol. 6, no. 2, pp. 163–168, 1963.
- [41] A. A. Goldstein and J. F. Price, "An effective algorithm for minimization," *Numer. Math.*, vol. 10, pp. 183–189, 1967.
- [42] A. A. Goldstein, *Constructive Real Analysis*. New York, NY, USA: Harper & Row, 1968.
- [43] C. G. Broyden, "The convergence of a class of double-rank minimization algorithms," *J. Inst. Math. Appl.*, vol. 6, no. 1, pp. 76–90, 1970.
- [44] R. Fletcher, "A new approach to variable metric algorithms," *Comput. J.*, vol. 13, no. 3, pp. 317–322, 1970.
- [45] D. Goldfarb, "A family of variable metric updates derived by variational means," *Math. Comput.*, vol. 24, no. 3, pp. 23–26, 1970.
- [46] D. F. Shanno, "Conditioning of quasi-newton methods for function minimization," *Math. Comput.*, vol. 24, no. 111, pp. 647–656, 1970.
- [47] *Pe-Expert4*, Myway, Inc., Jacksonville, FL, USA, Apr. 2019. [Online]. Available: https://www.myway.co.jp/en/products/pe_expert4.html
- [48] *Commercial Applications*, Harmonic Drive Systems, Inc., Tokyo, Japan, Jul. 2018. [Online]. Available: <http://www.hds.co.jp/english/products/example/>
- [49] Y. Sakagami, R. Watanabe, C. Aoyama, S. Matsunaga, N. Higaki, and K. Fujimura, "The intelligent ASIMO: System overview and integration," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2002, vol. 3, pp. 2478–2483.
- [50] Y. Kanai and Y. Fujimoto, "Torque-sensorless control for a powered exoskeleton using highly back-drivable actuators," in *Proc. 44th Annu. Conf. IEEE Ind. Electron. Soc.*, Washington, DC, USA, 2018, pp. 5116–5121, doi: 10.1109/IECON.2018.8591255.



Hiroshi Matsuki received the B.E., M.E., and Ph.D. degrees in electrical and computer engineering from Yokohama National University, Yokohama, Japan, in 2009, 2011, and 2014, respectively.

Since 2019, he has been with the Division of Systems and Information Engineering, Ashikaga University, Ashikaga, Japan. His research interests include machine learning, optimization problems, and artificial intelligence.



Kenta Nagano (S'15–M'18) received the B.E. degree in symbiotic systems science from Fukushima University, Fukushima, Japan, in 2013, and the M.E. and Ph.D. degrees in electrical and computer engineering from Yokohama National University, Yokohama, Japan, in 2015 and 2018, respectively.

His research interests include robotics, mechatronics, and motion control.



Yasutaka Fujimoto (S'93–M'98–SM'12) received the B.E., M.E., and Ph.D. degrees in electrical and computer engineering from Yokohama National University, Yokohama, Japan, in 1993, 1995, and 1998, respectively.

In 1998, he joined the Department of Electrical Engineering, Keio University, Yokohama, Japan. Since 1999, he has been with the Department of Electrical and Computer Engineering, Yokohama National University, where he is currently a Professor. His research interests include actuators, robotics, manufacturing automation, and motion control.

Dr. Fujimoto is an Associate Editor for the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and the *IEEJ Journal of Industry Applications*.